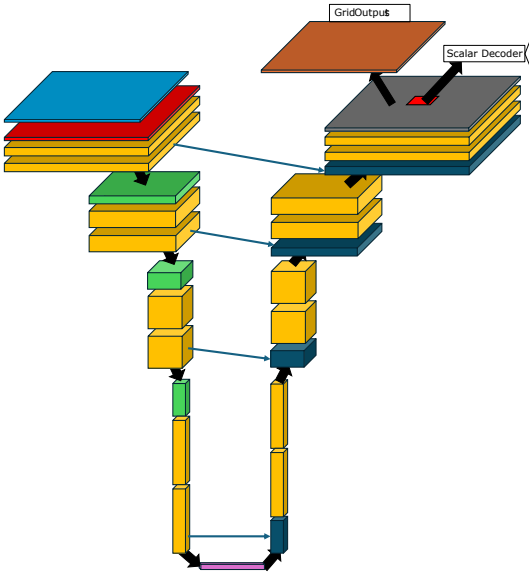
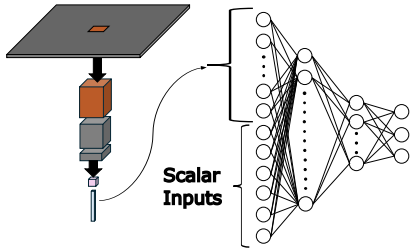
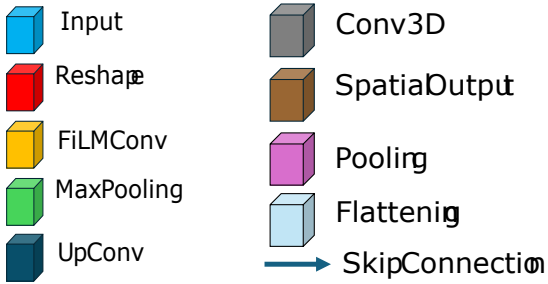






Block	Input Shape	Output Shape	Description
1	[2, 12, 61, 61]	[64, 16, 64, 64]	Padding, 2 FiLM Conv
2	[64, 16, 64, 64]	[128, 8, 32, 32]	Max Pooling, 2 FiLM Conv
3	[128, 8, 32, 32]	[256, 4, 16, 16]	Max Pooling, 2 FiLM Conv
4	[256, 4, 16, 16]	[512, 2, 8, 8]	Max Pooling, 2 FiLM Conv
5	[512, 2, 8, 8]	[1024, 1, 4, 4]	FiLM Conv
6	[1024, 1, 4, 4]	[512, 2, 8, 8]	UpConv, 2 FiLM Conv
7	[512, 2, 8, 8]	[256, 4, 16, 16]	UpConv, 2 FiLM Conv
8	[256, 4, 16, 16]	[128, 8, 32, 32]	UpConv, 2 FiLM Conv
9	[128, 8, 32, 32]	[64, 16, 64, 64]	UpConv, 2 FiLM Conv, Conv3d
10	[64, 16, 64, 64]	[2, 12, 61, 61]	Crop to Spatial Output dim
11	[64, 16, 64, 64]	[64, 12, 7, 7]	Crop to block around well
12	[64, 12, 7, 7]	[16, 12, 7, 7]	2 Conv3D
13	[16, 12, 7, 7]	[16, 2, 2, 2]	Pooling
14	[16, 2, 2, 2]	[128, 1]	Flattening to 1D array
15	[128, 1]	[32, 1]	3 Linear layers
16	[32, 1]	[1]	1 Linear layer: injection rate
17	[32, 1]	[1]	1 Linear layer: withdrawal rate
18	[32, 1]	[1]	1 Linear layer: hydrogen stored